transportation infrastructure often slow emergency healthcare response. Operating from strategically located distribution canters, Zipline uses fully autonomous fixed-wing drones capable of flying long distances to deliver essential medical products such as blood units, vaccines, antivenom, plasma, and emergency medica tions to remote clinics and district hospitals (Atiga, et al. 2024).

The drones follow GPS-guided flight paths and operate on a hub-andspoke distribution model. Supplies are packed into specialized temperature-controlled cargo bays to maintain medical integrity during flight. Upon reaching the destination, the drone does not land; instead, it performs a midair drop using biodegradable parachutes, ensuring rapid delivery even where landing infrastructure is absent. Once delivery is complete, the drone autonomously returns to the distribution canter for recovery, battery replacement, and reloading. This system drastically reduces travel time what traditionally required three to six hours by road can now be completed in 15 to 30 minutes, even in mountainous or flood-affected regions.

Zipline's operations demonstrate measurable impact on public health outcomes. Rwanda's national blood distribution efficiency improved significantly, with documented reductions in maternal mortality during childbirth, emergency trauma fatalities, and stock-out incidents in rural health centers. In Ghana, the Zipline network supports over 2000 health facilities, enabling continuous vaccine availability during the national immunization drive and COVID-19 pandemic. The program is coordinated through national health ministries, making it a flagship example of how UAV logistics can be institutionally integrated into public healthcare systems, rather than functioning as isolated pilot projects.